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Downregulation of Semaphorin 4A in keratinocytes reflects the features of non-lesional psoriasis

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Abstract

Psoriasis is a multifactorial disorder mediated by IL-17-producing T cells, involving immune cells and skin-constituting cells. Semaphorin 4A (Sema4A), an immune semaphorin, is known to take part in T helper type 1/17 differentiation and activation. However, Sema4A is also crucial for maintaining peripheral tissue homeostasis and its involvement in skin remains unknown. Here, we revealed that while Sema4A expression was pronounced in psoriatic blood lymphocytes and monocytes, it was downregulated in the keratinocytes of both psoriatic lesions and non-lesions compared to controls. Imiquimod application induced more severe dermatitis in Sema4A knockout (KO) mice compared to wild-type (WT) mice. The naïve skin of Sema4AKO mice showed increased T cell infiltration and IL-17A expression along with thicker epidermis and distinct cytokeratin expression compared to WT mice, which are hallmarks of psoriatic non-lesions. Analysis of bone marrow chimeric mice suggested that Sema4A expression in keratinocytes plays a regulatory role in imiquimod-induced dermatitis. The epidermis of psoriatic non-lesion and Sema4AKO mice demonstrated mTOR complex 1 upregulation, and the application of mTOR inhibitors reversed the skewed expression of cytokeratins in Sema4AKO mice. Conclusively, Sema4A-mediated signaling cascades can be triggers for psoriasis and targets in the treatment and prevention of psoriasis.

eLife

eLife assessment

This **important** study implicates Sempharon 4a in both mice and humans as a key suppressor of psoriatic inflammation. The data are in parts **incomplete** in defining the precise functionally relevant cellular source and mechanism. Nonetheless, this study brings new insight into psoriasis pathogenesis and a potential new therapeutic target.

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Introduction

While the infiltration of immune cells into skin plays a critical role in the development of psoriasis, as evidenced by interleukin (IL) -23/IL-17 axis (Fitch, Harper, Skorcheva, Kurtz, & Blauvelt, 2007; Hawkes, Yan, Chan, & Krueger, 2018; Kim & Krueger, 2017), recent studies have revealed that cells constructing skin structure, such as keratinocytes, fibroblasts, and endothelial cells, also play pivotal roles in the development (Heidenreich, Röcken, & Ghoreschi, 2009; Lowes, Russell, Martin, Towne, & Krueger, 2013; Zhang et al., 2023) and maintenance (Arasa et al., 2019; Francis et al., 2024; Li et al., 2023; Ma et al., 2023; Tan et al., 2015; Zhu et al., 2020) of psoriasis. Among these cells, keratinocytes function as a barrier and produce cytokines, chemokines, and antimicrobial peptides against foreign stimuli, resulting in the activation of immune cells (Ni & Lai, 2020; Zhou, Chen, Cui, Shi, & Guo, 2022).

Semaphorins were initially identified as guidance cues in neural development (Kolodkin, Matthes, & Goodman, 1993; Pasterkamp & Kolodkin, 2003) but are now regarded to play crucial roles in other physiological processes including angiogenesis (Iragavarapu-Charyulu, Wojcikiewicz, & Urdaneta, 2020; Serini, Maione, Giraudo, & Bussolino, 2009), tumor microenvironment (Hui, Tam, Jiao, & Ong, 2019; Jiang et al., 2022; Nakayama, Kusumoto, Nakahara, Fujiwara, & Higashiyama, 2018; Rajabinejad et al., 2020), and immune systems (Garcia, 2019; Kanth, Gairhe, & Torabi-Parizi, 2021; M. Naito & Kumanogoh, 2023). Semaphorin 4A (Sema4A), one of the immune semaphorins, plays both pathogenic and therapeutic roles in autoimmune diseases (Cavalcanti et al., 2020; He et al., 2023), allergic diseases (Maeda et al., 2019), and cancer (Iyer & Chapoval, 2018; Liu et al., 2018; Y. Naito et al., 2023; Pan, Wang, & Ye, 2016). While Sema4A expression on T cells is essential for T helper type 1 differentiation in the murine *Propionibacterium acnes*-induced inflammation model and delayed-type hypersensitivity model (Kumanogoh et al., 2005), Sema4A amplifies only T helper type 17 (Th17)-mediated inflammation in the effector phase of murine experimental autoimmune encephalomyelitis (Koda et al., 2020). Accordingly, in multiple sclerosis, high serum Sema4A levels correlate with the elevated serum IL-17A, earlier disease onset, and increased disease severity (Koda et al., 2020; Nakatsuji et al., 2012). In anti-tumor immunity, research involving human samples and murine models suggests that Sema4A expressed in cancer cells and regulatory T cells promotes tumor progression (Delgoffe et al., 2013; Liu et al., 2018; Pan et al., 2016), while other reports reveal that Sema4A in cancer cells and dendritic cells bolsters anti-tumor immunity by enhancing CD8 T cell activity (Y. Naito et al., 2023; Suga et al., 2021). In addition to these roles in immune reactions, mice with a point mutation in Sema4A develop retinal degeneration, suggesting that Sema4A is also crucial for peripheral tissue homeostasis (Nojima et al., 2013). These reports suggest that the role of Sema4A can differ based on the disease, phase, and involved cells.

Herein, we investigated the roles of Sema4A in the pathogenesis of psoriasis by analyzing skin and blood specimens from psoriatic subjects and using a murine model.

Results

Epidermal Sema4A expression is downregulated in psoriasis

The analysis of previously-published single-cell RNA-sequencing data from control (Ctl) and psoriatic lesional (L) skin specimens (Kim et al., 2023) revealed detectable expression of *SEMA4A* in keratinocytes, dendritic cells, and macrophages in both Ctl and L. *SEMA4A* expression was low in neural crest-like cells, fibroblasts, CD4 T cells, CD8 T cells, NK cells, and plasma cells, making the comparison of expression levels impractical (**Figure 1A** and **B**; **Figure 1-figure supplement 1A-C**). While dendritic cells and macrophages showed comparable *SEMA4A* expression levels between Ctl and L, expression in keratinocytes was significantly downregulated in L (**Figure 1C**).

Immunohistochemistry of Ctl and psoriasis demonstrated Sema4A expression in keratinocytes (**Figure 1D**). The staining intensity of Sema4A in epidermis was significantly lower in both non-lesions (NL) and L than in Ctl (**Figure 1E**). Relative mRNA expression of *SEMA4A* was also decreased in the epidermis of NL and L compared to Ctl, while it remained comparable in the dermis (**Figure 1F**). In contrast, the proportions of Sema4A-positive cells were significantly higher in blood CD4 and CD8 T cells, and monocytes in psoriasis compared to Ctl (**Figure 1G**; **Figure 1-figure supplement 2**). Serum Sema4A levels, measured by enzyme-linked immunosorbent assay (ELISA), were comparable between Ctl and psoriasis (**Figure 1H**). These findings demonstrate that the expression profile of Sema4A in psoriasis varies across cell types.

Psoriasis-like dermatitis is augmented in Sema4AKO mice

When psoriasis-like dermatitis was induced in wild-type (WT) mice and Sema4A knockout (KO) mice by imiquimod application on ears (**Figure 2A**), ear swelling on day 4 was more pronounced in Sema4AKO mice (**Figure 2B**) with upregulated *Il17a* gene expression (**Figure 2C**). Flow cytometry analysis of cells isolated from the ears revealed increased proportions of $V\gamma2^+$ T cells, $V\gamma2^-V\gamma3^-$ double-negative (DN) γδ T cells, and IL-17A-producing cells of those fractions in Sema4AKO epidermis (**Figure 2D** and **E**; Figure 2-figure supplement 1). In Sema4AKO dermis, there was also an increase in the proportions of $V\gamma2^+$ T cells and IL-17A-producing $V\gamma2^+$ T cells (**Figure 2D** and **E**). These results suggest that Sema4A deficiency in mice accelerates psoriatic profile. IL-17A-producing T cells in skin-draining lymph nodes (dLN) remained comparable between WT mice and Sema4AKO mice (**Figure 2F**).

Sema4A in keratinocytes may play a role in preventing murine psoriasis-like dermatitis

To investigate the cells responsible for the augmented ear swelling in Sema4AKO mice, bone marrow chimeric mice were next analyzed (**Figure 3A**). Mice deficient in non-hematopoietic Sema4A (WT→KO) displayed more pronounced ear swelling than mice with intact Sema4A expression (WT→WT) following imiquimod application (**Figure 3B**). Similarly, mice with a systemic deficiency of Sema4A (KO→KO) showed severe ear swelling compared to mice deficient in hematopoietic Sema4A (KO→WT) (**Figure 3B**). Ear swelling was comparable between WT→WT mice and KO→WT mice (**Figure 3B**). Flow cytometry analysis revealed increased infiltration of IL-17A-producing DN γδ T cells in the epidermis, as well as $V\gamma2^+$ T cells and IL-17A-producing $V\gamma2^+$ T cells in the dermis, in WT→KO mice compared to WT→WT mice (**Figure 3C** and **D**). These findings suggest that non-hematopoietic cells, possibly keratinocytes, are primarily responsible for the increased imiquimod-induced Sema4AKO mice ear swelling.

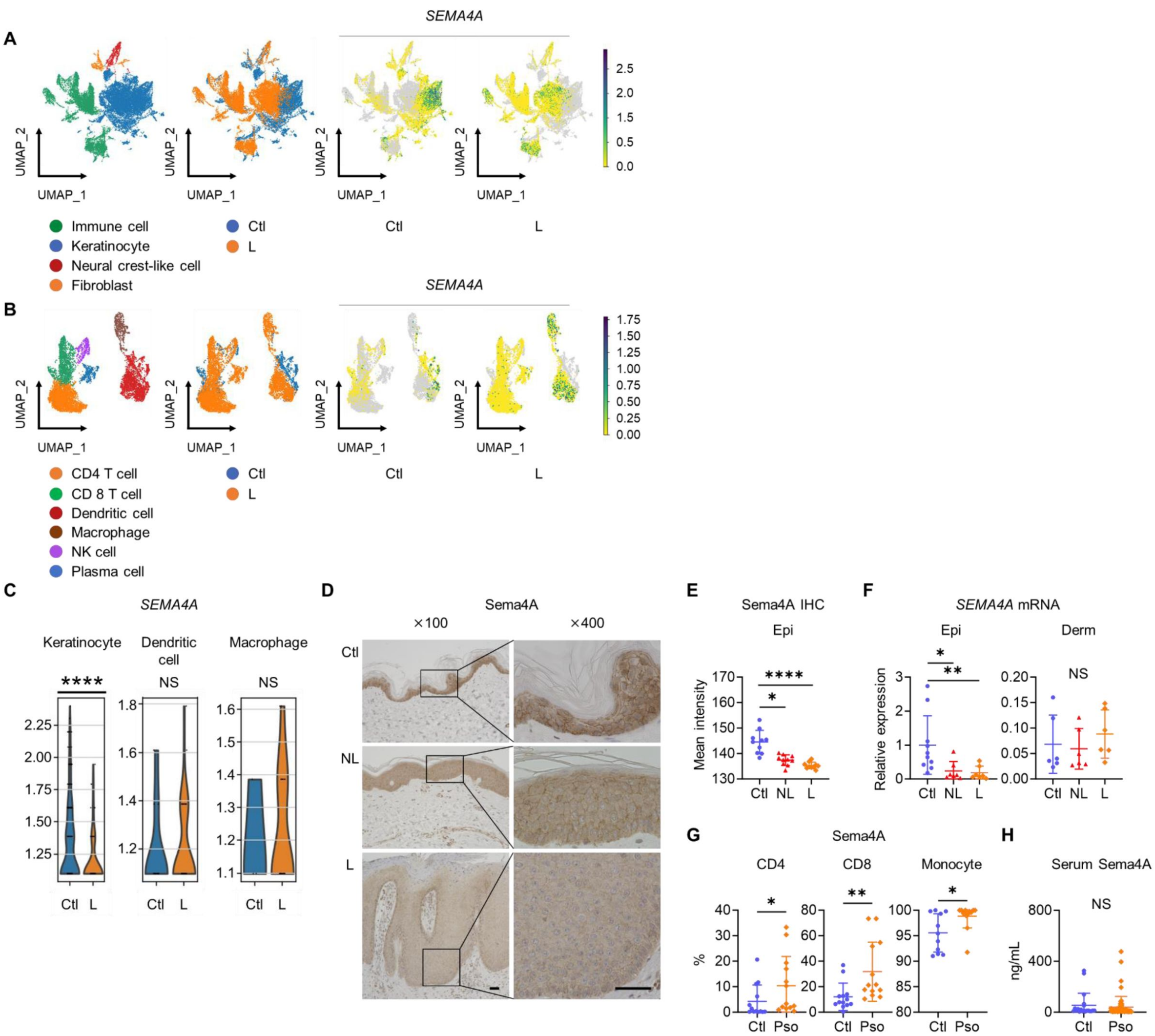

Figure 1

Epidermal Sema4A expression is downregulated in psoriasis.

(**A**) UMAP plots, generated from single-cell RNA-sequencing data (GSE220116), illustrate cell distributions from control (Ctl) and psoriatic lesion (L) samples (*n* = 10 for Ctl, *n* = 11 for L). (**B**) Subclustering of immune cells. (**C**) *SEMA4A* expression in keratinocytes, dendritic cells, and macrophages. ****$padj$ < 0.001. NS, not significant. Analyzed using Python and cellxgene VIP. (**D**) Representative immunohistochemistry and magnified views showing Sema4A expression in Ctl, psoriatic non-lesion (NL), and L. Scale bar = 50 μm. (**E**) Mean epidermal (Epi) Sema4A intensity in immunohistochemistry (*n* = 10 per group). Each dot represents the average intensity from 5 unit areas per sample. (**F**) Relative *SEMA4A* expression in Epi (*n* = 10 for Ctl, L, *n* = 7 for L and NL) and dermis (Derm, *n* = 6 per group). (**G**) Proportions of Sema4A-expressing cells in blood CD4 T cells (left), CD8 T cells (middle), and monocytes (right) from Ctl and psoriatic (Pso) patients. (*n* = 13 per group in CD4 and CD8, *n* = 11 for Ctl and *n* = 13 for Pso in Monocytes). (**H**) Serum Sema4A levels in Ctl (*n* = 20) and Pso (*n* = 60). **E** to **H**: *p < 0.05, **p < 0.01, ****p < 0.0001. NS, not significant.

Figure1-source data1

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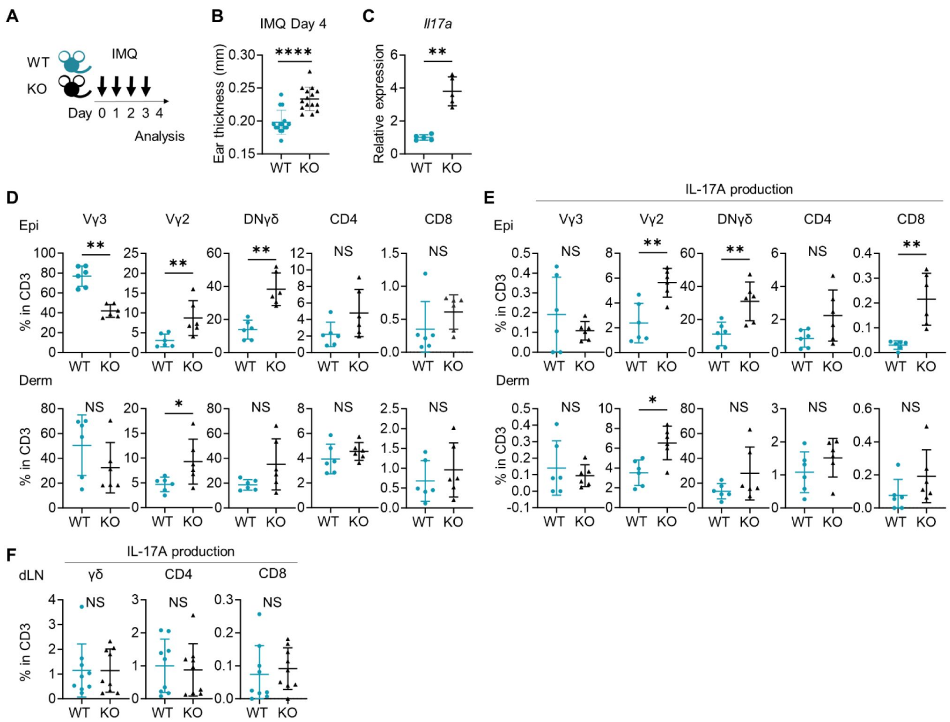

Figure 2

Psoriasis-like dermatitis is augmented in Sema4AKO mice.

(**A**) Experimental scheme. Wild-type (WT, green) mice and Sema4A knockout (KO, black) mice were treated with 10 mg/ear of 5% Imiquimod (IMQ) for 4 consecutive days. Samples for flow cytometry analysis were collected on Day 4. (**B**) Ear thickness of WT mice and KO mice on Day 4 (n = 15 per group). (**C**) Relative expression of *Il17a* in epidermis (n = 5 per group). (**D** and **E**) The percentages of Vγ3, Vγ2, Vγ2-Vγ3-γδ (DNγδ), CD4, and CD8 T cells (**D**) and those with IL-17A production (**E**) in CD3 fraction in the Epi (top) and Derm (bottom) of WT and KO ears (n = 6 per group, each dot represents the average of 4 ear specimens). (**F**) The percentages of IL-17A-producing γδ, CD4, and CD8 T cells in CD3 fraction in skin-draining lymph nodes (dLN) (n = 9 per group). **B**-**F**: *$p < 0.05$, **$p < 0.01$, ****$p < 0.0001$. NS, not significant.

Figure2-source data1

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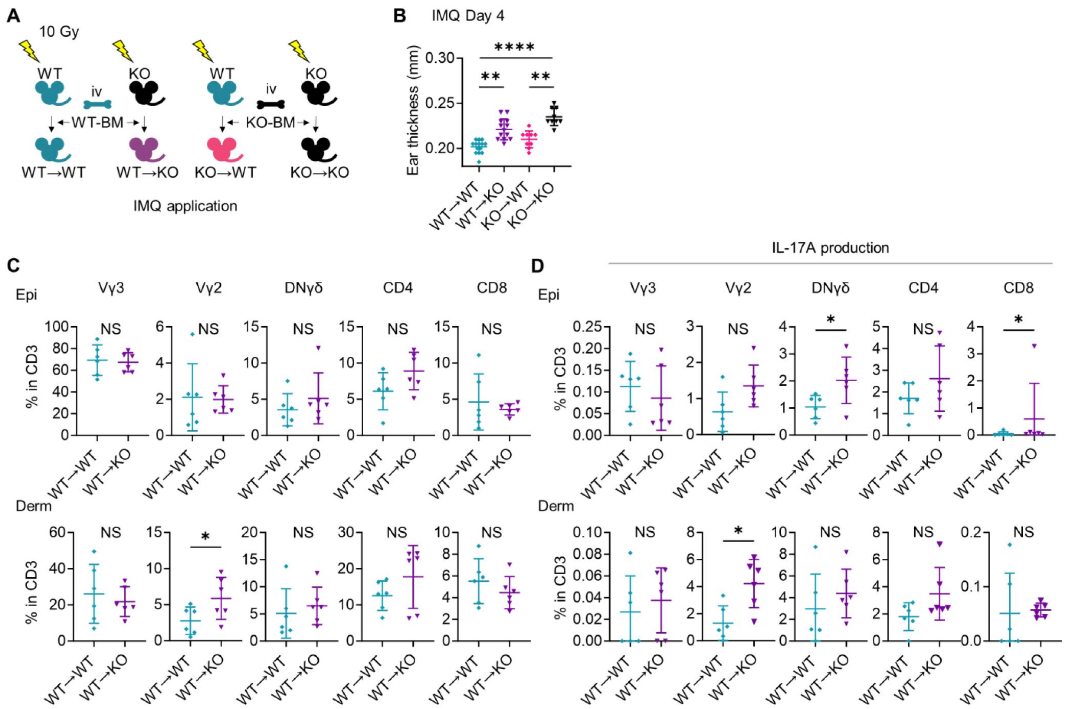

Figure 3

Sema4A in keratinocytes may play a role in preventing murine psoriasis-like dermatitis.

(**A**) Experimental scheme for establishing BM chimeric mice. (**B**) Day 4 ear thickness in the mice with the indicated genotypes (n = 14 for WT→WT, n = 13 for WT→KO, n =9 for KO→WT, n = 9 for KO→KO). (**C** and **D**) The percentages of Vγ3, Vγ2, DNγδ, CD4, and CD8 T cells (**C**) and IL-17A production (**D**) in CD3 fraction from Day 4 Epi (top) and Derm (bottom) of the ears from WT→ WT mice and WT→ KO mice (n = 6 per group, each dot represents the average of 4 ear specimens). **B**-**D**: *p < 0.05, **p < 0.01, ****p < 0.0001. NS, not significant.

Figure3-source data1

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Sema4AKO epidermis is thicker than WT epidermis with increased γδ T17 infiltration

Even without imiquimod application, Sema4AKO ears turned out to be slightly but significantly thicker than WT ears on week 8 while their appearance remained normal (**Figure 4A**). While epidermal thickness of back skin was comparable at birth (**Figure 4B**), on week 8, ear epidermis of Sema4AKO mice was notably thicker than that of WT mice (**Figure 4B**), suggesting that acanthosis in Sema4AKO mice is accentuated post-birth. Dermal thickness remained comparable between WT mice and Sema4AKO mice at both times (**Figure 4B**). WT epidermis showed significantly higher *Sema4a* mRNA expression compared to the WT dermis (**Figure 4C**). Based on these observations, Sema4A appears to play a more pronounced role in epidermis than in dermis.

Sema4AKO epidermis exhibited increased expression of *Ccl20, Tnfa*, and *Il17a* and a trend of upregulation of *S100a8* compared to WT epidermis (**Figure 4D**). These differences were not observed in dermis (**Figure 4D**). Flow cytometry analysis revealed increased infiltration of γδ T cells in Sema4AKO ear (**Figure 4E**). These cells predominantly expressed resident memory T cell (T_{RM})-characteristic molecules, CD69 and CD103 (**Figure 4E**). Sema4AKO skin also had a higher number of T_{RM} in both CD4 and CD8 T cells (**Figure 4E**). The percentages of Vγ2$^+$ T cells, DNγδ T cells, and CD8 T cells in epidermis were higher in Sema4AKO mice than in WT mice, which was not the case in dermis (**Figure 4F**). The proportion of Vγ3$^+$ dendritic epidermal T cells was comparable between WT mice and Sema4AKO mice (**Figure 4F**). Epidermal Vγ2$^+$ T cells and DNγδ T cells in Sema4AKO mice showed higher IL-17A-producing capability (**Figure 4G**), while IFNγ and IL-4 production was comparable between WT mice and Sema4AKO mice (**Figure 4-figure supplement 1A** and **B**). Conversely, the frequency of IL-17A-producing T cells from dLN was comparable (**Figure 4-figure supplement 1C**). The production of IL-17A and IFNγ from splenic T cells under T17-polarizing conditions remained consistent between WT mice and Sema4AKO mice (**Figure 4-figure supplement 2**).

Taken together, it is suggested that T17 cells are specifically upregulated in epidermis, indicating that the epidermal microenvironment plays a pivotal role in facilitating the increased T cell infiltration observed in naïve Sema4AKO mice.

Sema4AKO skin shares features with human psoriatic NL

Previous literatures have identified certain features common to psoriatic L and NL, such as thickened epidermis (**Figure 5-figure supplement 1**) (Gallais Sérézal et al., 2019), CCL20 upregulation (Gallais Sérézal et al., 2019), and accumulation of IL-17A-producing T cells (Vo et al., 2019), which were detected in Sema4AKO mice.

Gene expression analysis using public RNA-sequencing data (Tsoi et al., 2019) with RaNA-seq (Prieto & Barrios, 2019) showed upregulation of keratinization and antimicrobial peptide genes in NL compared to Ctl (**Figure 5A**). Gene Ontology analysis highlighted an upregulation in biological processes, predominantly in peptide cross-linking involved in epidermis formation and keratinocyte differentiation, with a secondary increase in the defense response to viruses in NL (**Figure 5B**). While the expression of *Keratin (KRT) 10* was comparable between NL and Ctl, upregulation in *KRT5*, *KRT14*, and *KRT16* was observed in NL (**Figure 5C**).

In the murine model, relative expression levels of *Krt10*, *Krt14*, *Krt16*, and *filaggrin* were elevated in Sema4AKO epidermis. (**Figure 6A**). Immunofluorescence analysis showed that Sema4AKO epidermis had a higher density of keratinocytes positive for Krt5, Krt10, Krt14, and Krt16

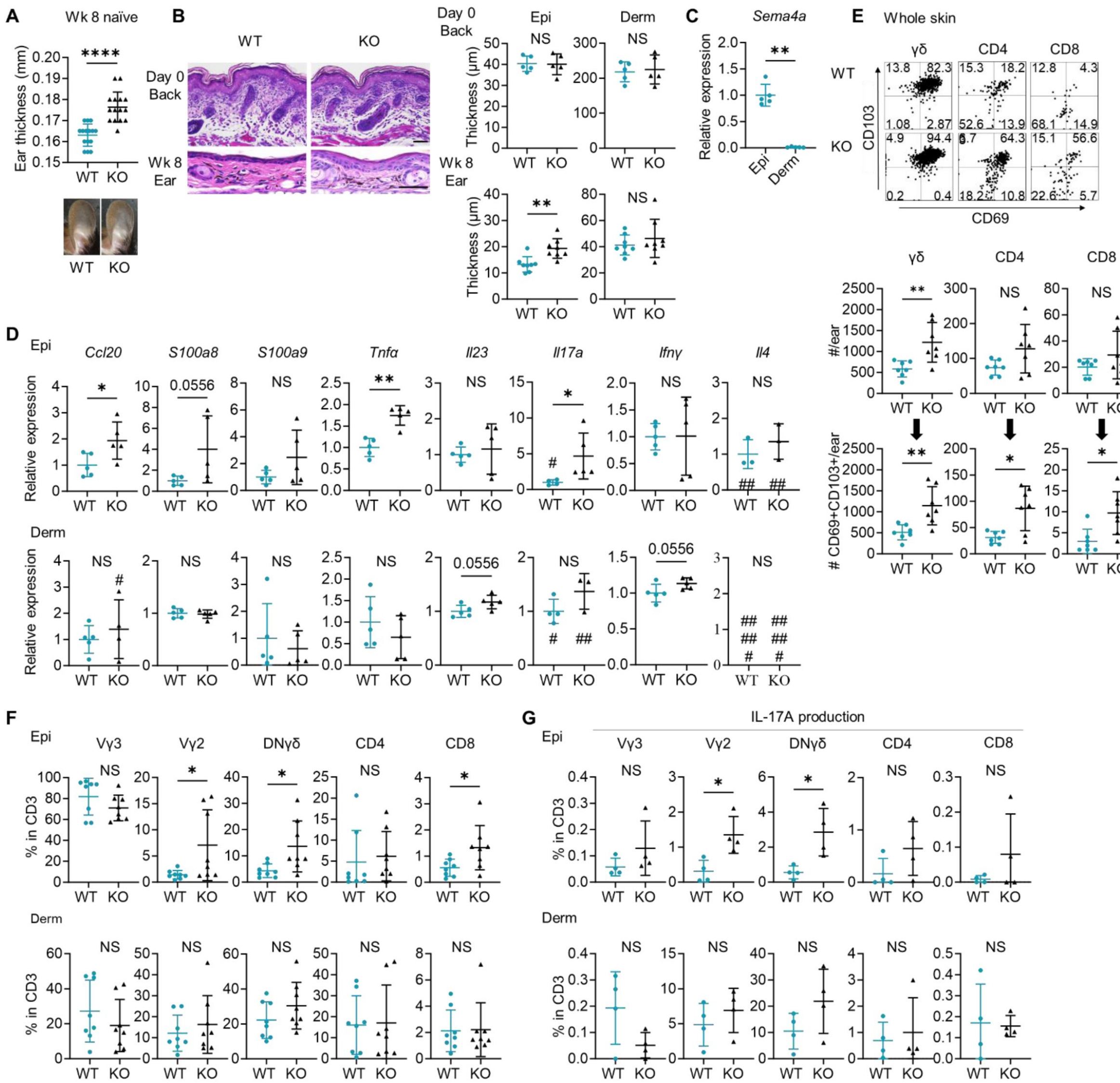

Figure 4

Naïve Sema4AKO epidermis is thicker than WT epidermis with increased γδ T17 infiltration.

(**A**) Ear thickness of WT mice and KO mice at week (wk) 8 (*n* = 15 per group) and representative images. (**B**) Left: representative hematoxylin and eosin staining of day 0 back and wk 8 ear. Scale bar = 50 µm. Right: Epi and Derm thickness in d 0 back (*n* = 5) and wk 8 ear (*n* = 8). (**C**) Relative *Sema4a* expression in WT Epi and Derm (*n* = 5 per group). (**D**) Relative expression of psoriasis-associated genes in Epi (top) and Derm (bottom) of WT mice and KO mice (*n* = 5 per group, #: not detected). (**E**) Representative dot plots showing CD69 and CD103 expression in the indicated T cell fractions from whole skin. The graphs show T cell counts per ear (top) and those with resident memory phenotype (bottom) (*n* = 7 per group). (**F** and **G**) The percentages of the indicated T cell fractions (**F**, *n* = 8 per group) and IL-17A production (**G**, *n* = 4 per group) in Epi (top) and Derm (bottom). **E**-**G**: Each dot represents the average of 4 ear specimens. **A**-**G**: *$p < 0.05$, **$p < 0.01$, ****$p < 0.0001$. NS, not significant.

Figure4-source data1

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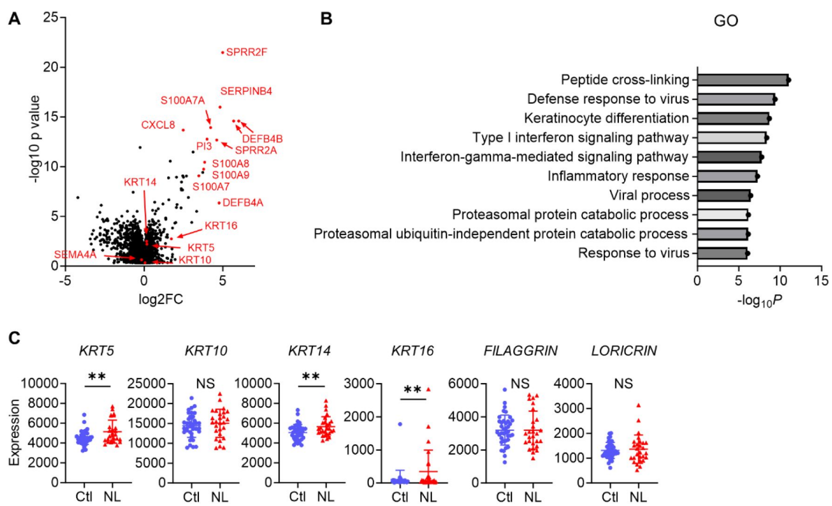

Figure 5

Psoriatic NL shows aberrant differentiation of keratinocytes.

(**A** and **B**) The volcano plot (**A**) and Gene ontology (GO) analysis (**B**), generated from RNA-sequencing data (GSE121212) using RaNA-seq, display changes in gene expression in psoriatic NL compared to Ctl. (**C**) The difference in the expression of epidermal differentiation markers between Ctl and NL (n = 38 for Ctl, n = 27 for NL) was calculated with the transcripts per million values. ***padj* < 0.01. NS, not significant.

Figure5-source data1

Excel file containing quantitative data for **Figure 5**.

compared to WT epidermis (**Figure 6B**). This upregulation was not observed in back skin at birth (**Figure 6-figure supplement 1**). Comparable transepidermal water loss between WT mice and Sema4AKO mice indicated preserved skin barrier function in Sema4AKO mice (**Figure 6C**).

Based on these results, it is implied that the epidermis of human psoriatic NL and Sema4AKO mice exhibit shared pathways, potentially leading to an acanthotic state. Combined with the observed acanthosis and increased T17 infiltration in Sema4AKO mice, Sema4AKO skin is regarded to demonstrate the features characteristic of human psoriatic NL.

mTOR signaling is upregulated in the epidermis of psoriatic NL and Sema4AKO mice

Previous reports have shown that mTOR pathway plays a critical role in maintaining epidermal homeostasis, as evidenced by mice with keratinocyte-specific deficiencies in Mtor, Raptor, or Rictor, which exhibit a hypoplastic epidermis with impaired differentiation and barrier formation (Asrani et al., 2017; Ding et al., 2016; Ding et al., 2020). We thus investigated mTOR pathway in both human and murine epidermis.

Immunohistochemical analyses highlighted the increase in phospho-S6 (p-S6), indicating the upregulation of mTOR complex (C) 1 signaling, in the epidermal upper layers of both L and NL compared to Ctl (**Figure 7A**). The activation of mTORC2 in the epidermis, inferred from phospho-Akt (p-Akt), was scarcely detectable in L, NL, and Ctl (**Figure 7A**). In the epidermis of WT mice and Sema4AKO mice, the upregulation of both mTORC1 and mTORC2 signaling was observed in Sema4AKO epidermis by immunohistochemistry (**Figure 7B**). Western blot analysis showed upregulated mTORC1 signaling in Sema4AKO epidermis compared to WT epidermis (**Figure 7C**). The enhancement of mTORC1 and mTORC2 signaling became obvious in the Sema4AKO epidermis after developing psoriatic dermatitis by imiquimod application (**Figure 7D**).

Inhibition of mTOR signaling modulates cytokeratin expression in Sema4AKO mice

To investigate the contribution of mTORC1 and mTORC2 signaling in the development of psoriatic features in the Sema4AKO epidermis, mTORC1 inhibitor rapamycin and mTORC2 inhibitor JR-AB2-011 were intraperitoneally applied to Sema4AKO mice for 14 days. Although epidermal thickness remained unchanged by the inhibitors (**Figure 8A** and **B**), relative gene expression of *Krt5* was significantly upregulated and that of *Krt16* was significantly downregulated after rapamycin application (**Figure 8C**). While the upregulation of *Il17a* in Sema4AKO epidermis was not clearly modified by rapamycin (**Figure 8C**), immunofluorescence revealed the decrease in the number of CD3 T cells in Sema4AKO epidermis by rapamycin (**Figure 8D**). We additionally conducted topical application of rapamycin gel and vehicle gel on the left and right ears of Sema4AKO mice, respectively. Although there were no detectable changes in epidermal thickness and epidermal T cell counts, the upregulation of *Krt5* and downregulation of *Krt16* was observed again (**Figure 8-figure supplement 1**). Conversely, the application of JR-AB2-011 resulted in decreased expression of *Krt5*, *Krt10*, and *Krt14* with a trend towards increased *Krt16* expression (**Figure 8E**). JR-AB2-011 did not influence the number of infiltrating T cells in the epidermis (**Figure 8F**). mTORC1 primarily regulates keratinocyte proliferation, whereas mTORC2 mainly involved in the keratinocyte differentiation through Sema4A-related signaling pathways. The observed similarities in *Il17a* expression following treatment with rapamycin and JR-AB2-011 suggest that the production of these cytokines is not significantly dependent on Sema4A-related mTOR signaling.

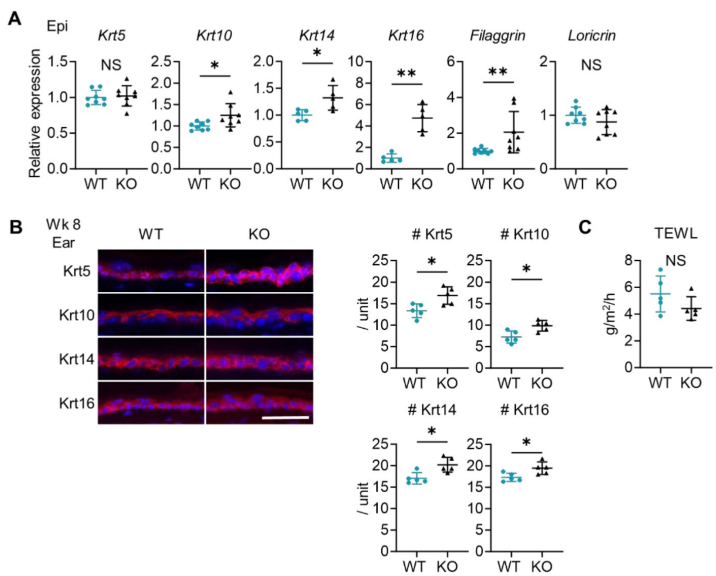

Figure 6

Sema4AKO skin shares the features of human psoriatic NL.

(**A**) Relative gene expression of epidermal differentiation markers between wk 8 Epi of WT mice and KO mice (n = 5 for *Krt14* and *Krt16*, n = 8 for *Krt5*, *Krt10*, *Filaggrin,* and *Loricrin*). (**B**) Left: Representative immunofluorescence pictures of Krt5, Krt10, Krt14, and Krt16 (red) overlapped with DAPI. Scale bar = 50 µm. Right: Accumulated graphs showing the numbers of Krt5, Krt10, Krt14, and Krt16 positive cells per 100 µm width (n = 5 per group) of wk 8 ear (right). Each dot represents the average from 5 unit areas per sample. (**C**) Transepidermal water loss (TEWL) in back skin of WT mice and KO mice at wk 8 (n = 5 per group). **A**-**C**: *$p < 0.05$, **$p < 0.01$. NS, not significant.
Figure 6-source data1
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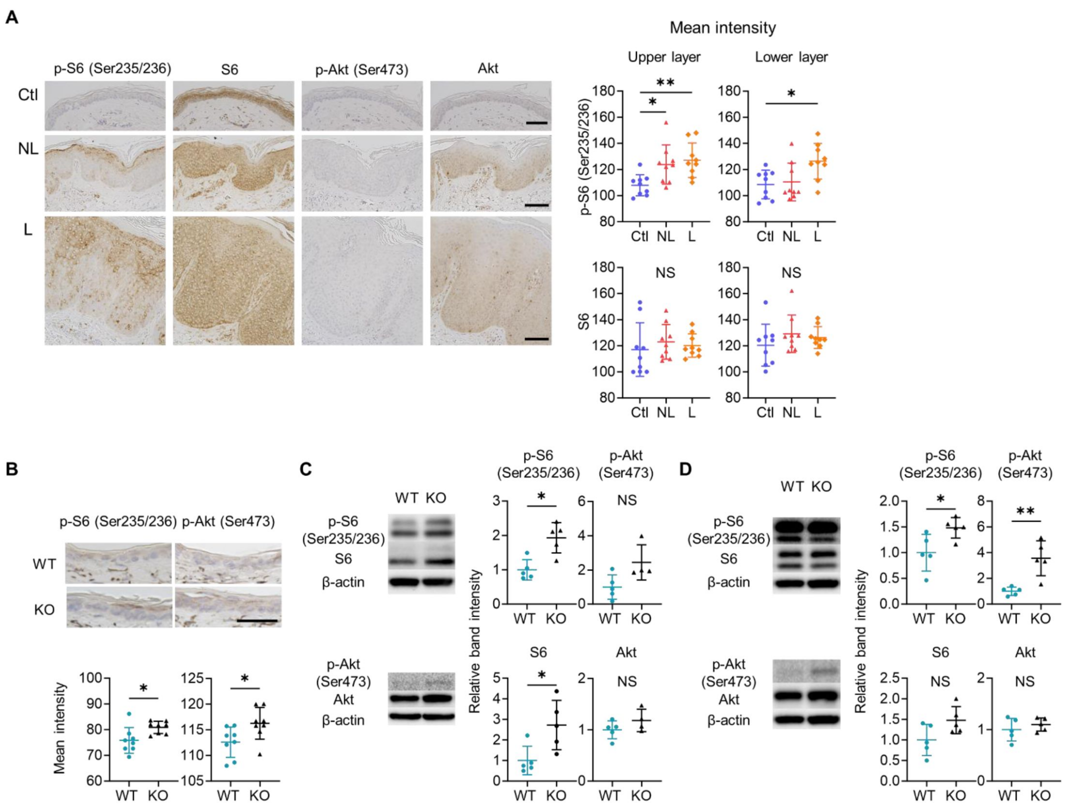

Figure 7

mTOR signaling is upregulated in the epidermis of psoriatic NL and Sema4AKO mice.

(**A**) Representative results of immunohistochemistry displaying cells positive for p-S6 (Ser235/236), S6, p-Akt (Ser473), and Akt in Ctl, NL, and L. The graphs of accumulated data show the mean intensity of p-S6 and S6 in the upper and lower epidermal layers (n = 9 per group). Scale bar = 100 μm. Each dot represents the average mean intensity from 5 unit areas per sample. (**B**) The mean intensity of p-S6 (Ser235/236) and p-Akt (Ser473), detected by immunohistochemistry in the epidermis of WT mice and KO mice, were analyzed. Scale bar = 50 μm. Each dot represents the average intensity from 5 unit areas per sample (n = 8 per group). (**C** and **D**) Immunoblotting of p-S6 (Ser235/236), S6, p-Akt (Ser473), and Akt in tissue lysates from epidermis without treatment (**C**) and with IMQ treatment for consecutive 4 days (**D**) (n = 5 per group, except for p-Akt and Akt in **C**, for which n = 4). **A**-**D**: *p < 0.05, **p < 0.01. NS, not significant.

Figure 7-source data1

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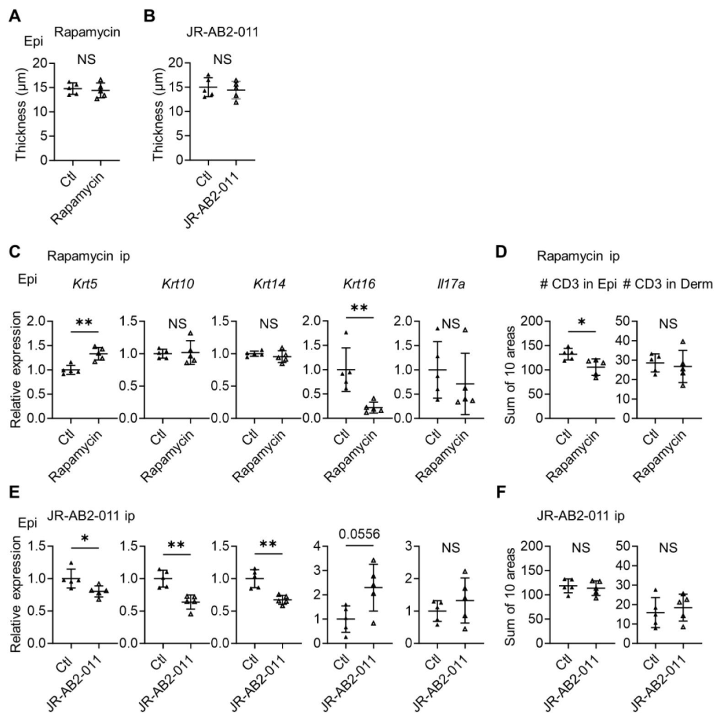

Figure 8

Inhibitors of mTOR signaling modulate the expression of cytokeratins in Sema4AKO mice.

(**A** and **B**) Epidermal thickness of Sema4AKO mice treated intraperitoneally with vehicle (Ctl) or rapamycin (**A**), and Ctl or JR-AB2-011 (**B**) (n = 5 per group). (**C** and **D**) Relative expression of keratinocyte differentiation markers and *Il17a* in Sema4AKO Epi (**C**), and the number of T cells in Epi and Derm under Ctl or rapamycin (**D**) (n = 5 per group). (**E** and **F**) Relative expression of keratinocyte differentiation markers and *Il17a* in Sema4AKO Epi (**E**), and the number of T cells in Epi and Derm under Ctl or JR-AB2-011 (**F**) (n = 5 per group). **D** and **F**: Each dot represents the sum of numbers from 10 unit areas across 3 specimens. **A**-**F**: *$p < 0.05$, **$p < 0.01$. NS, not significant.

Figure 8-source data1

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Discussion

Recent studies have highlighted the significance of IL-17A-producing T_{RM} in psoriatic NL (Cheuk et al., 2014; Gallais Sérézal et al., 2018; Vo et al., 2019; Vu, Koguchi-Yoshioka, & Watanabe, 2021). This condition is also characterized by epidermal thickening with elevated CCL20 expression (Gallais Sérézal et al., 2019) and a distinct cytokeratin expression pattern, marked by increased levels of Krt5, Krt14, and Krt16 (Tsoi et al., 2019) (**Figure 5C**). Here, we identified that murine Sema4A deficiency could induce these features of psoriatic NL.

Sema4A expression in epidermis, but not in dermis, was diminished in psoriatic L and NL compared to Ctl. While Sema4A expression on blood immune cells was upregulated in psoriasis, its serum levels were comparable between Ctl and psoriasis. Despite the reported involvement of increased Sema4A expression on immune cells in the pathogenesis of multiple sclerosis (Koda et al., 2020; Nakatsuji et al., 2012), our results suggest that the diminished expression of Sema4A in skin-constructing cells plays a more prominent role in the pathogenesis of psoriasis than its increased expression on immune cells.

In our murine model, regardless of imiquimod application, IL-17A-producing T cells were increased in Sema4AKO skin although their frequency was comparable in dLN of WT mice and Sema4AKO mice. Additionally, the potential for *in vitro* T17 differentiation did not differ between T cells from WT mice and Sema4AKO mice. These findings suggest that the absence of Sema4A in the skin microenvironment plays a crucial role in the localized upregulation of IL-17A-producing T cells in Sema4AKO mice.

It is well-documented that upregulated mTORC1 signaling promotes the pathogenesis of psoriasis (Buerger, 2018; Karagianni, Pavlidis, Malakou, Piperi, & Papadavid, 2022; Ruf, Andreoli, Itin, Pluschke, & Schmid, 2014), and that rapamycin can partially ameliorate the disease activity in psoriatic subjects and murine models (Bürger et al., 2017; Gao & Si, 2018; Reitamo et al., 2001). Our results using Sema4AKO mice revealed that the inhibition of mTORC1 leads to the downregulation of *Krt16* and that of mTORC2 leads to the downregulation of undifferentiated keratinocytes, while the Sema4AKO epidermal thickening and the upregulated IL-17A signaling is not reversed by mTOR blockade. It is plausible that the downregulation of Sema4A can lead to the upregulation of mTORC1 and mTORC2 signaling in keratinocytes, and the augmented signaling leads to the psoriatic profile of proliferation and differentiation, which is a part of the psoriatic NL disposition.

This study has limitations. Sema4A expression in skin cells other than keratinocytes was not thoroughly investigated. However, since single-cell RNA-sequencing has shown that Sema4A is predominantly expressed in keratinocytes, dendritic cells, and macrophages, with a notable reduction in keratinocytes in psoriasis, we infer that keratinocytes are the primary cells responsible for the psoriatic features resulting from Sema4A downregulation. We were not able to determine whether Sema4A functions as a ligand or a receptor in epidermis (Ito & Kumanogoh, 2016; Kumanogoh et al., 2002; Lu et al., 2018; Sun et al., 2017) in this study, either. Although both mTORC1 and mTORC2 signals are upregulated in Sema4AKO epidermis, we were not able to confirm mTORC2 signaling from human skin due to technical limitation and sample limitation, while the results from augmented mTORC2 signal in Sema4AKO mice and the normalization of Krt5 and Krt14 by mTORC2 inhibitor imply the involvement of mTORC2 signal in psoriatic epidermis. The role of Sema4A other than mTOR signaling, which may be involved in the regulation of T17 induction in skin, is not discussed, either. These limitations should be overcome in the near future.

In summary, epidermal Sema4A downregulation can reflect the psoriatic non-lesional features. Thus, targeting the downregulated Sema4A and upregulated mTOR signaling in psoriatic epidermis can be a promising strategy for psoriasis therapy and modification of psoriatic diathesis in NL for the prevention of disease development.

Materials and methods

Human sample collection

Psoriatic L and NL skin specimens were acquired from 17 psoriasis patients. Ctl specimens were obtained from 19 subjects who underwent tumor resection or reconstructive surgery. For epidermal-dermal separation, specimens were incubated overnight at 4℃ with 2.5 mg/mL dispase (Wako) in IMDM (Wako). Blood samples were collected from 73 psoriasis and 33 Ctl. In addition to ELISA (Nakatsuji et al., 2012), peripheral blood mononuclear cells were isolated using Ficoll-Paque PLUS density gradient media (Cytiva). Patient details are provided in **Supplemental Table 1**.

Mice

C57BL/6J WT mice were procured from CLEA Japan. Semaphorin 4A KO mice with C57BL/6J background were generated as previously described (Kumanogoh et al., 2005). We examined female mice in order to reduce the result variation. Neonatal mice and female mice aged 8 to 12 weeks, maintained under specific pathogen-free conditions, were used.

To develop psoriasis-like dermatitis, 10 mg of 5% imiquimod cream (Mochida) was applied to both ears for 4 consecutive days. Ear thickness was measured using a thickness gauge (Peacock).

For bone marrow reconstitution, 2×10^5 bone marrow cells obtained from the indicated strains were intravenously transferred to recipient mice that had received a single 10 Gy irradiation. The reconstituted mice were subjected to the experiments in 8 weeks.

In the specified experiments, skin specimens were separated into epidermis and dermis by 5 mg/mL dispase (Wako) in IMDM for 30 minutes at 37℃.

Transepidermal water loss measurements were performed on the back skin of mice at week 8 using a Tewameter (Courage and Khazaka Electronic GmbH), according to the manufacturer's instructions.

Immunofluorescence and immunohistochemistry

Specimens were fixed in 4% paraformaldehyde phosphate buffer solution (Wako), embedded in paraffin, and sliced into 3-μm thickness on glass slides. After deparaffinization and rehydration, antigen retrieval was performed using citrate buffer (pH 6.0, Nacalai tesque) or TE buffer (pH 9.0, Nacalai tesque).

For immunohistochemistry, samples were incubated with 3% H_2O_2 (Wako) for 5 minutes. After blocking (Agilent), the specimens were incubated with the indicated primary antibodies under specified conditions (**Supplemental Table 2**). The specimens were applied with the EnVision Detection Systems Peroxidase/DAB, Rabbit/Mouse (Agilent) and counterstained with hematoxylin (Wako). For immunofluorescence, the blocked specimens were incubated with the indicated primary antibodies followed by the secondary antibodies (Table E4). Mounting media with DAPI (VectaShield) was used. Slides were observed using a fluorescence microscope (BZ-700, Keyence). The staining intensity was measured using Fiji software (ImageJ, National Institutes of Health) over lengths of 200 μm in human samples and 100 μm in murine samples. This measurement was taken from five areas of each specimen, and the average score is presented. The average thickness of epidermis and dermis from 10 spots is presented. The number of epidermal cells positive for the indicated cytokeratins was counted per 100 μm width, with the average number from 5 areas being presented. Additionally, the numbers of CD3-positive cells in murine ears were counted across 10 fields of 400 μm width, with the total sum being presented.

Quantitative reverse transcription polymerase chain reaction (qRT-PCR)

Total RNA was extracted from homogenized skin tissue using Direct-zol RNA MiniPrep Kit (ZYMO Research). cDNA was synthesized by High-Capacity RNA-to-cDNATM Kit (Thermo Fisher Scientific). qPCR was performed using TB Green Premix Ex Taq II (Takara-bio) on a ViiA 7 Real-Time PCR System (Thermo Fisher Scientific). The primers are listed in **Supplemental Table 3**. Relative gene expression levels were normalized to the housekeeping gene GAPDH using the ΔΔCT technique.

Flow cytometry analysis

Single-cell suspensions from murine skin-dLN and spleen were prepared by grinding, filtering, and lysing red blood cells (BioLegend). Skin specimens were minced and digested with 3 mg/mL collagenase III (Worthington Biochemical Corporation) in RPMI 1640 medium (Wako) at 37°C for 10 minutes for epidermis, and 30 minutes for dermis or whole skin. Cells were surface-stained with directly conjugated monoclonal antibodies (**Supplemental Table 4**). Dead cells were identified using LIVE/DEAD^{TM} Fixable Dead Cell Stain Kit (Thermo Fisher Scientific). To evaluate cytokine production, cells were stimulated with PMA (50 ng/mL, Wako) and ionomycin (1000 ng/mL, Wako), plus Golgiplug (BD Biosciences) for 4 hours before surface staining. Fixation, permeabilization, and intracellular cytokine staining were performed using BD Cytofix/Cytoperm^{TM} Fixation/Permeabilization Kit (BD Biosciences) according to the manufacturer's protocol. In specified experiments, the numbers of each cell subset per ear were estimated using CountBright Absolute Counting Beads (Thermo Fisher Scientific). Sample analysis was conducted using BD FACSCanto II (BD Biosciences), and data were analyzed with Kaluza software (Beckman Coulter).

In vitro Th17 differentiation

Murine splenic T cells were isolated using Pan T Cell Isolation Kit II (Miltenyi Biotec). Two hundred thousand T cells per well were cultured in 96-well plates in the presence of T Cell Activation/Expansion Kit (Miltenyi Biotec) for 2 weeks. The medium was supplemented twice per week with the following recombinant cytokines: mouse recombinant IL-6 (20 ng/mL), IL-1β (20 ng/mL), and IL-23 (40 ng/mL) for the IL-23 dependent Th17 cell condition; IL-6 (20 ng/mL) and TGFβ1 (3 ng/mL) for the IL-23 independent Th17 cell condition; or IL-23 (40 ng/mL) for the IL-23 only Th17 cell condition. The cytokines listed were purchased from BioLegend (**Supplemental Table 5**). Afterward, the cultured cells were processed for flow cytometry analysis.

Western blotting

Murine epidermis was lysed using RIPA buffer (Wako) containing proteinase- and protease-inhibitor cocktail (Nacalai tesque). Protein lysates were separated by 10% SuperSepTM Ace (Wako), transferred onto polyvinylidene difluoride membranes (0.45 μm, Merck) by Trans-Blot Turbo Transfer System (Bio-Rad). The membrane was blocked with 5% bovine serum albumin and subjected to immunoblotting targeting the indicated proteins overnight 4°C, followed by the application of HRP-conjugated secondary antibody for one hour at room temperature (**Supplemental Table 2**). WB stripping solution (Nacalai tesque) was used to remove the antibodies for further evaluation.

Inhibition of mTOR

Rapamycin (4 mg/kg, Sanxin Chempharma), JR-AB2-011 (400 μg/kg, MedChemExpress) and vehicle were applied intraperitoneally to mice once daily for 14 consecutive days. Rapamycin and JR-AB2-011 were dissolved in 100% DMSO and diluted with 40% Polyethylene Glycol 300 (Wako), 5% Tween-80 (Sigma-Aldrich) and 45% saline in sequence. For topical application, 0.2% rapamycin gel and vehicle gel were prepared by the pharmaceutical department as previously described (Wataya-Kaneda et al., 2017). Rapamycin gel was applied to the left ear, and vehicle gel to the right ear (10mg/ear) of Sema4AKO, once daily for 14 consecutive days.

Data processing of single-cell RNA-sequencing and bulk RNA-sequencing

The raw count matrix data from the previously reported single-cell RNA-sequencing data from GSE220116 (Kim et al., 2023) were imported into Scanpy (1.9.6) using Python for further analyses. For each sample, cells and genes meeting the following criteria were excluded: cells expressing over 200 genes (sc.pp.filter_cells), cells with a high proportion of mitochondrial genes (> 5%), and genes expressed in fewer than 3 cells (sc.pp.filter_genes). Counts were normalized using sc.pp.normalize_per_cell, logarithmized (sc.pp.log1p), and scaled (sc.pp.scale). Highly variable genes were selected using sc.pp.filter_genes_dispersion with the options min_mean = 0.0125, max_mean = 2.5, and min_disp = 0.7.

Principal Component analysis was conducted with sc.pp.pca, selecting the 1st to 50th Principal Components for embedding and clustering. Neighbors were calculated with batch effect correction by BBKNN (Polański et al., 2020). Embedding was performed by sc.tl.umap, and cells were clustered using sc.tl.leiden. Some cells were further subclassified in a similar manner. Differentially expressed genes were identified on the cellxgene VIP, with significance defined as *padj* < 0.05.

Bulk RNA-sequencing data from GSE121212 (Tsoi et al., 2019) were re-analyzed using RaNAseq (Prieto & Barrios, 2019) for differential expression in Ctl versus psoriatic NL. Our analysis included normalization, differential gene expression (defining significance as *padj* < 0.05), and focused on Gene Ontology biological process analysis. The gene expression was calculated with the transcripts per million values.

Statistical analysis

GraphPad Prism 10 software (GraphPad Software) was used for all statistical analyses except for RNA-sequencing. Mann-Whitney test was used for two-group comparisons, while Kruskal-Wallis test followed by Dunn's multiple comparisons test was applied for comparison among three or more groups. Statistical significance was defined as $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.001$ (***). In the analysis of RNA-sequencing data, statistical significance was defined as *padj* < 0.01 (**), and *padj* < 0.001 (***).

Study approval

All experiments involving human specimens were in accordance with the Declaration of Helsinki and were approved by the Institutional Review Board in Osaka University Hospital (20158-6). Written informed consent was obtained from all participants. All murine experiments were approved by Osaka University Animal Experiment Committee (J007591-013) and all procedures were conducted in compliance with the Guidelines for Animal Experimentation established by Japanese Association for Laboratory Animal Science.

Data availability

The single-cell RNA-sequencing datasets generated by Kim et al. (Kim et al., 2023) and Tsoi et al. (Tsoi et al., 2019) used in this study are available in the NCBI Gene Expression Omnibus under accession codes GSE220116 and GSE121212, respectively. Values for all data points in graphs are reported in the Source data.

Acknowledgements

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Supplementary materials

A

Clinical information					
Sample name	Title	Age	Gender	PASI	Site
Sample01	Control01	37	Male	0	Trunk
Sample02	Control02	37	Male	0	Trunk
Sample03	Control03	66	Male	0	Trunk
Sample04	Control04	40	Male	0	Leg (right)
Sample05	Control05	40	Male	0	Leg (right)
Sample06	Control06	41	Male	0	Leg (right)
Sample07	Control07	52	Female	0	Arm (left)
Sample08	Control08	52	Female	0	Arm (left)
Sample09	Control09	74	Male	0	Trunk
Sample10	Control10	74	Male	0	Trunk
Sample11	Psoriasis01_preTx_LS	52	Female	18	Arm (right)
Sample12	Psoriasis02_preTx_LS	63	Male	7.8	Back
Sample13	Psoriasis03_preTx_LS	58	Female	8.4	Leg (left)
Sample15	Psoriasis04_preTx_LS	41	Male	9.6	Trunk
Sample17	Psoriasis05_preTx_LS	46	Female	6.8	Arm (right)
Sample18	Psoriasis06_preTx_LS	39	Male	6	Leg (right)
Sample21	Psoriasis07_preTx_LS	39	Male	9.8	Trunk
Sample22	Psoriasis07_preTx_LS_02	40	Male	18	Trunk
Sample26	Psoriasis08_preTx_LS	48	Female	7.4	Arm (right)
Sample27	Psoriasis09_preTx_LS	45	Male	6.4	Leg (right)
Sample28	Psoriasis10_preTx_LS	38	Male	8.4	Leg (right)

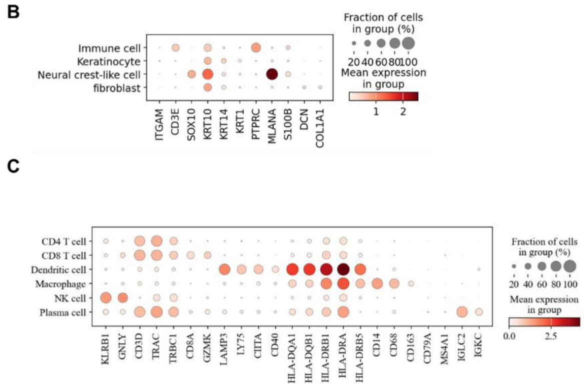

Figure 1-figure supplement 1

Sema4A is downregulated in the keratinocytes of lesional psoriasis in the single-cell RNA-sequencing data.

(**A**) Sample information for specimens from Ctl and psoriatic L (GSE220116). (**B** and **C**) Clusters of cells were identified by their expression patterns of signature genes.

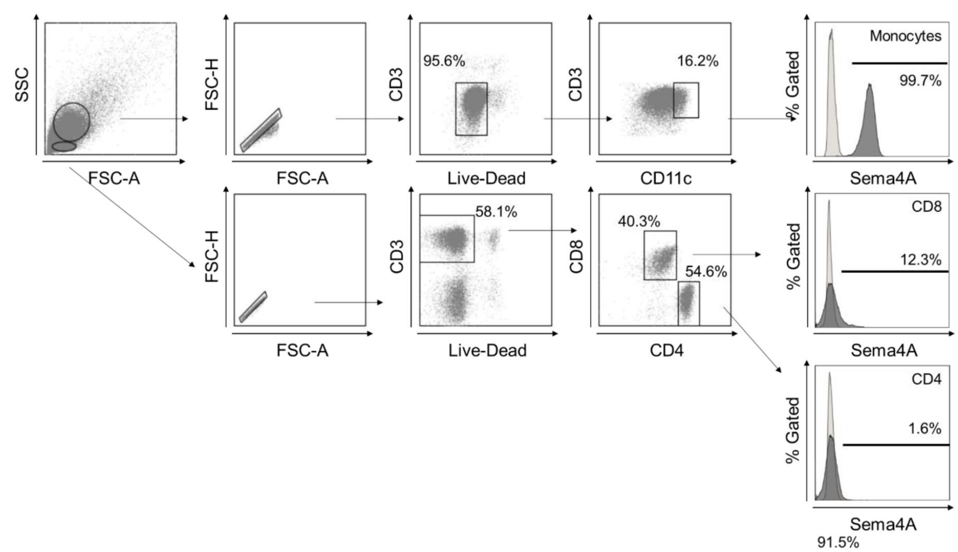

Figure 1-figure supplement 2

Gating strategy in flow cytometry.

Gating strategy for human Sema4A expression in blood cells. Large and small cells were distinguished using forward scatter (FCS) and side scatter (SSC) in a dot plot panel, with dead cells being excluded. Monocytes were defined within the live large cell population as CD11c positive. CD4 and CD8 T cells were identified within the live small cell population as CD3 positive CD4 positive and CD3 positive CD8 positive populations, respectively. The empty histogram represents the flow cytometry minus one control for Sema4A.

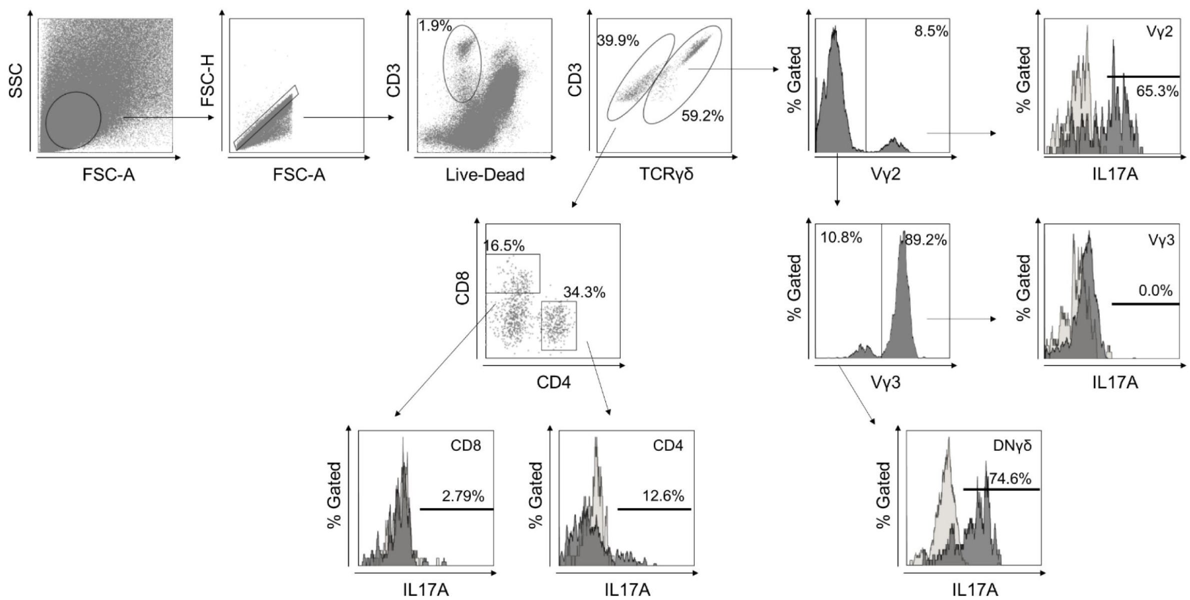

Figure 2-figure supplement 2

Gating strategy in flow cytometry.

Gating strategy for murine T cells infiltrating the epidermis and dermis. After excluding dead cells, TCRγδ positive T cells were evaluated for the expression of Vγ2. TCRγδ positive Vγ2 negative population was further assessed the expression of Vγ3. The CD3 positive TCRγδ negative population was evaluated for the expression of CD4 and CD8. Each population was analyzed for cytokine production. The empty histogram represents the isotype control for IL-17A.

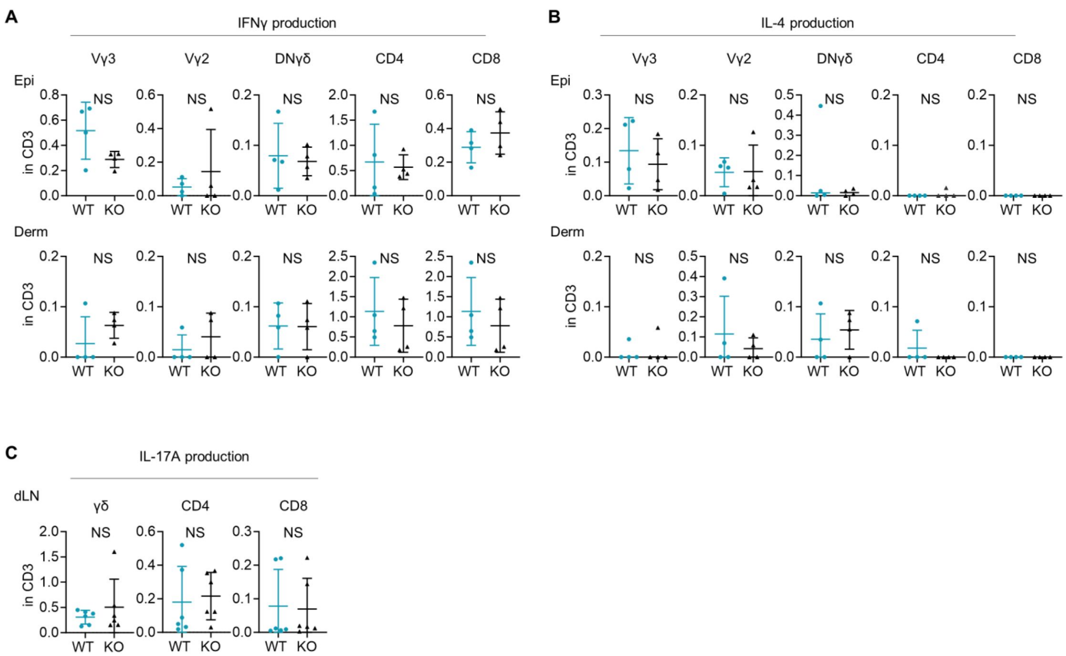

Figure 4-figure supplement 1

Expression of IFNγ and IL-4 is comparable between WT and Sema4AKO skin.

(**A** and **B**) The graphs presenting the percentages of IFNγ (**A**) and IL-4 (**B**) -producing Vγ2, Vγ2-Vγ3-γδ (DNγδT), CD4, and CD8 T cells in CD3 fraction in the Epi (top) and Derm (bottom) of WT mice and Sema4AKO mice (n = 4 per group, each dot represents the average of 4 ear specimens). (**C**) The graphs showing the percentages IL-17A-producing γδ, CD4, and CD8 T cells in CD3 fraction from draining LN (dLN) of WT mice and Sema4AKO mice (n = 6 per group). **A**-**C**: NS, not significant.
Figure 4-figure supplement1-source data1
Excel file containing quantitative data for **Figure 4-figure supplement1**.

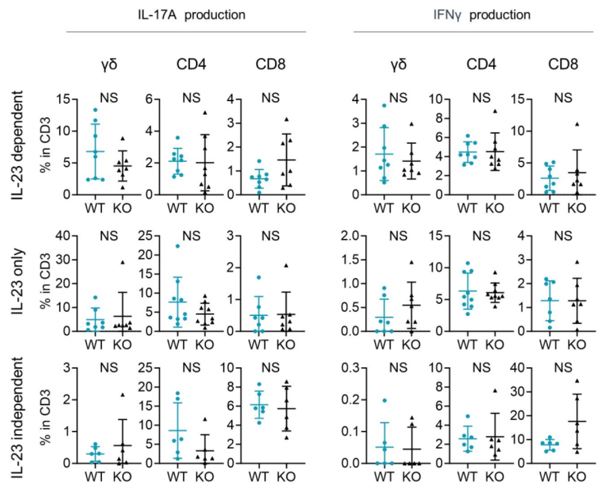

Figure 4-figure supplement 2

Comparable T17 differentiation potential under Th17-skewing conditions between WT mice and Sema4AKO mice.

Splenic T cells were cultured for 2 weeks, followed by flow cytometry analysis. The accumulated data display the percentages of IL-17A-producing (right) and IFNγ-producing (left) γδ, CD4, and CD8 T cells within CD3 fraction under various conditions: IL-23 dependent Th17-skewing condition (top), IL-23 only Th17-skewing condition (middle), and IL-23 independent Th17-skewing condition (bottom) . NS, not significant.

Figure 4-figure supplement2-source data1

Excel file containing quantitative data for **Figure 4-figure supplement2**.

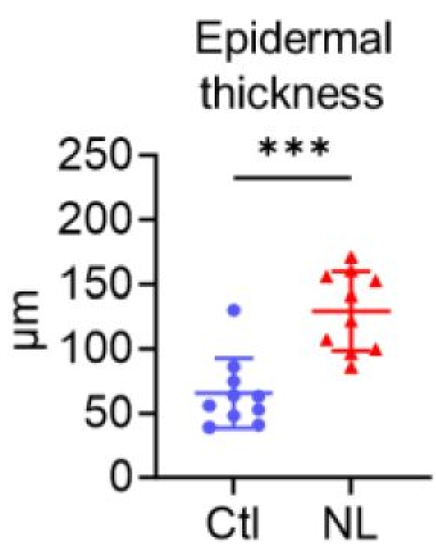

Figure 5-figure supplement 1

The epidermis of psoriatic non-lesion is thicker than that of control skin.

Epidermal thickness of Ctl and psoriatic NL (n = 10 per group). ***$p < 0.001$.

Figure 5-figure supplement1-source data1

Excel file containing quantitative data for **Figure 5-figure supplement1**.

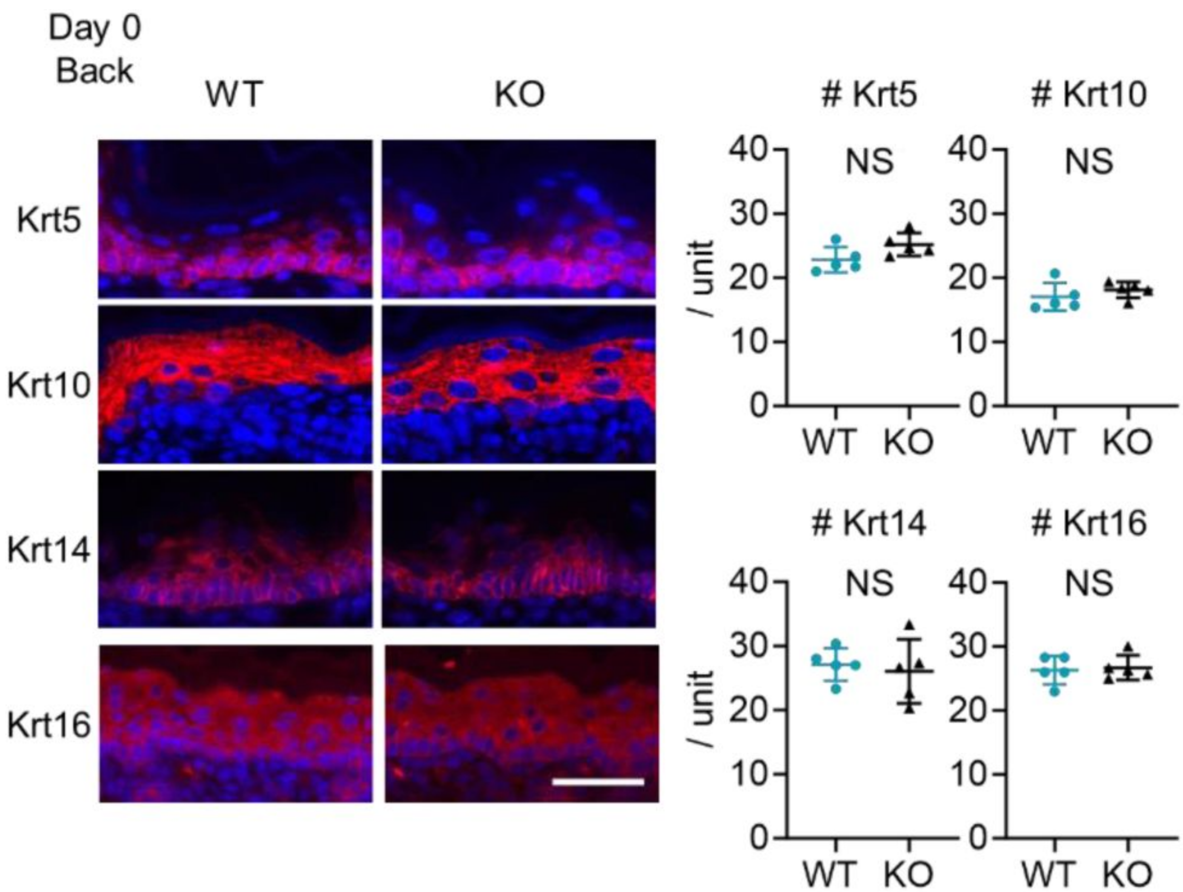

Figure 6-figure supplement 1

Upregulation of cytokeratin expression related to psoriasis is not detected at birth in Sema4AKO mice.

Representative immunofluorescence pictures of Krt5, Krt10, Krt14, and Krt16 (red) overlapped with DAPI, and the accumulated graphs showing the numbers of Krt5, Krt10, Krt14, and Krt16 positive cells per 100 μm width (n = 5 per group) in the epidermis of day 0 back. Scale bar = 50 μm. Each dot represents the average from 5 unit areas per sample. NS, not significant.

Figure 6-figure supplement1-source data1

Excel file containing quantitative data for **Figure 6-figure supplement1**.

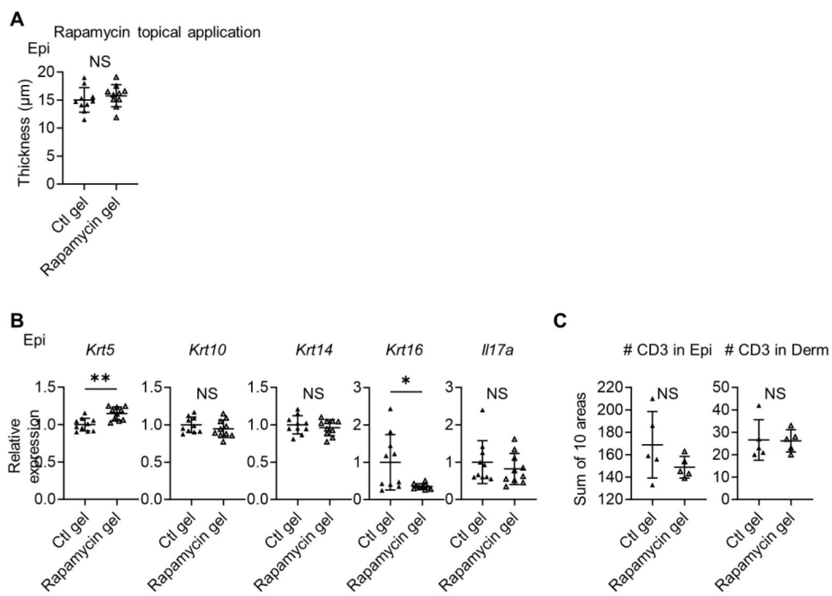

Figure 8-figure supplement 1

Topical application of rapamycin gel yields partially similar results to intraperitoneal treatment.

(**A**) Comparison of epidermal thickness between vehicle (Ctl) gel-treated right ears and rapamycin gel-treated left ears of Sema4AKO mice (n = 10 per group). (**B**) Relative expression of keratinocyte differentiation markers and *Il17a* in Sema4AKO Epi under Ctl gel or rapamycin gel treatments (n = 5 per group). (**C**) The number of T cells in the epidermis (left) and dermis (right), under Ctl gel or rapamycin gel treatments (n = 5 per group). Each dot represents the sum of numbers from 10 unit areas across 3 specimens. **A**-**C**: *$p < 0.05$, **$p < 0.01$. NS, not significant.

Figure 8-figure supplement1-source data1

Excel file containing quantitative data for **Figure 8-figure supplement1**.

Psoriasis severity is defined by the total body surface area (BSA) affected: <3% BSA for mild, 3%-10% BSA for moderate, and >10% BSA for severe disease.

Fig. 1, D and E — Analysis of Sema4A expression by immunohistochemistry

Fig. S4 — Epidermal thickness

Psoriasis (n = 10)

Age	Gender	Body site	Severity
66	Male	Leg	Severe
55	Male	Trunk	Moderate
80	Female	Trunk	Severe
72	Male	Trunk	Moderate
30	Male	Trunk	Severe
72	Female	Leg	Moderate
63	Male	Arm	Mild
38	Male	Trunk	Mild
35	Male	Leg	Mild
71	Male	Trunk	Moderate

Control (n = 10)

Age	Gender	Body site
76	Male	Neck
52	Female	Leg
66	Female	Leg
33	Male	Leg
68	Female	Trunk
57	Female	Leg
65	Male	Trunk
41	Female	Trunk
51	Female	Trunk
66	Female	Trunk

Fig 1F — Analysis of Sema4A expression by qRT-PCR

Epidermis

Psoriasis (n = 7)

Age	Gender	Body site	Severity
71	Male	Leg	Moderate
76	Female	Neck	Mild
38	Female	Trunk	Moderate
49	Female	Trunk	Moderate
61	Male	Arm	Mild
68	Female	Trunk	Moderate
72	Male	Leg	Mild

Control (n = 10)

Age	Gender	Body site
57	Female	Leg
69	Female	Trunk

Table S1

Patient information

66	Female	Trunk
86	Female	Trunk
53	Female	Trunk
84	Male	Neck
37	Female	Trunk
46	Female	Trunk
51	Female	Trunk
46	Female	Trunk

Dermis

Psoriasis (n = 6)

Age	Gender	Body site	Severity
76	Female	Neck	Mild
38	Female	Trunk	Moderate
49	Female	Trunk	Moderate
61	Male	Arm	Severe
68	Female	Trunk	Moderate
72	Male	Leg	Severe

Control (n = 6)

Age	Gender	Body site
69	Female	Trunk
86	Female	Trunk
84	Male	Neck
37	Female	Trunk
51	Female	Trunk
46	Female	Trunk

Fig 1G

Analysis of Sema4A expression by flow cytometry

	Cell type	Average of age	Male (n =)	Female (n =)
Psoriasis (n = 13)	CD4 T cell, CD8 T cell, and monocyte	56.1	9	4
Control (n = 13)	CD4 T cell and CD8 T cell	52.2	6	7
Control (n = 11)	Monocyte	54.1	6	5

Fig 1H

Analysis of Serum Sema4A by ELISA

	Average of age	Male (n =)	Female (n =)
Psoriasis (n = 60)	55.3	43	17
Control (n = 20)	52.1	11	9

Fig 6A

Pictures of p-S6, S6, p-Akt, and Akt staining

Psoriasis (n = 1)

Age	Gender	Body site	Severity
30	Male	Trunk	Severe

Control (n = 1)

Age	Gender	Body site
66	Female	Trunk

Table S1 (continued)

Analysis of p-S6, S6 expression by immunohistochemistry

Psoriasis (n = 9)

Age	Gender	Body site	Severity
66	Male	Leg	Severe
55	Male	Trunk	Moderate
80	Female	Trunk	Severe
72	Male	Trunk	Moderate
30	Male	Trunk	Severe
72	Female	Leg	Moderate
38	Male	Trunk	Mild
35	Male	Leg	Mild
71	Male	Trunk	Moderate

Control (n = 9)

Age	Gender	Body site
76	Male	Neck
66	Female	Leg
33	Male	Leg
68	Female	Trunk
57	Female	Leg
65	Male	Trunk
41	Female	Trunk
51	Female	Trunk
66	Female	Trunk

Table S1 (continued)

Target	Clone	Species	Supplier	Catalog	Dilution	Incubation time	Incubation temperature
Sema4A	Polyclonal	Rabbit	Abcom	ab70178	100	30 min	RT
phospho-S6 (Ser235/236)	D57.2.2E	Rabbit	Cell Signaling	4858	400	overnight	4 °C
S6	5G10	Rabbit	Cell Signaling	2217	100	overnight	4 °C
phospho-Akt (Ser473)	D9E	Rabbit	Cell Signaling	4060	100	overnight	4 °C
Akt	C67E7	Rabbit	Cell Signaling	4691	300	overnight	4 °C
Keratin 5	Polyclonal	Rabbit	BioLegend	905503	800	overnight	4 °C
Keratin 10	Polyclonal	Rabbit	BioLegend	905403	400	overnight	4 °C
Keratin 14	Polyclonal	Rabbit	BioLegend	905303	400	overnight	4 °C
Cytokeratin 16	8L6R4	Rabbit	Invitrogen	MA5-42892	100	overnight	4 °C
CD3	CD3-12	Rat	BIO-RAD	MCA1477	100	overnight	4 °C
Rabbit IgG H&L (Alexa Fluor 555)	Polyclonal	Donkey	Abcam	ab150074	1000	30 min	RT
Rat IgG H&L (Alexa Fluor 555)	Polyclonal	Donkey	Abcam	ab150154	1000	30 min	RT
Rabbit IgG H&L (Alexa Fluor 488)	Polyclonal	Goat	ThermoFisher	A-11034	500	30 min	RT
β-actin	AC-15	Mouse	Sigma-Aldrich	A5441	5000	overnight	4 °C
Anti-Mouse IgG, HRP-Linked Whole Ab	-	Sheep	Cytiva	NA931	10000	1 hour	RT
Anti-Rabbit IgG, HRP-Linked Whole Ab	-	Donkey	Cytiva	NA934	10000	1 hour	RT

RT: Room temperature

Table S2

Antibodies used for immunohistochemical, immunofluorescence, and western blot analyses

Gene Symbol	Forward primer (5'-3')	Reverse primer (5'-3')
GAPDH	GTCTCCTCTGACTTCAACAGCG	ACCACCCTGTTGCTGTAGCCAA
SEMA4A	TCTGCTCCTGAGTGGTGATG	AAACCAGGACACGGATGAAG

Primer sequences of real-time quantitative PCR in murine sample experiments

Gene Symbol	Forward primer (5'-3')	Reverse primer (5'-3')
Gapdh	TGTGTCCGTCGTGGATCTGA	TTGCTGTTGAAGTCGCAGGAG
Krt5	CAGAGCTGAGGAACATGCAG	CATTCTCAGCCGTGGTACG
Krt10	CGTACTGTTCAGGGTCTGGAG	GCTTCCAGCGATTGTTTCA
Krt14	CGTACTGTTCAGGGTCTGGAG	GCTTCCAGCGATTGTTTCA
Krt16	TGAGCTGACCCTGTCCAGA	TGAGCTGACCCTGTCCAGA
Filaggrin	GGAGGCATGGTGGAACTGA	TGTTTATCTTTTCCCTCACTTCTACATC
Loricrin	TCACTCATCTTCCCTGGTGCTT	GTCTTTCCACAACCCACAGGA
Sema4a	ATGGCCCTACCATCCCTGG	AGCAGCGTGTCAAAGTCTCG
Ccl20	ATGGCCTGCGGTGGCAAGCGT	CATCTTCTTGACTCTTAGGC
S100a8	TGCGATGGTGATAAAAGTGG	GGCCAGAAGCTCTGCTACTC
S100a9	CACCCTGAGCAAGAAGGAAT	TGTCATTTATGAGGGCTTCATTT
Ifgγ	ATCTGGAGGAACTGGCAAAA	TTCAAGACTTCAAAGAGTCTGAGGTA
Tnfα	GCCTCCCTCTCATCAGTTCT	CACTTGGTGGTTTGCTACGA
Il4	ACAGGAGAAGGGACGCCAT	GAAGCCCTACAGACGAGCTCA
Il17a	CTGTGTCTCTGATGCTGTTG	ATGTGGTGGTCCAGCTTTC
Il23	TCCCTACTAGGACTCAGCCAAC	GCTGCCACTGCTGACTAGAA

Table S3

Primer sequences of real-time quantitative PCR used in human sample experiments

Target	Clone	Species	Supplier	Catalog
CD3	OKT3	Mouse	BioLegend	317335
CD4	RPA-T4	Mouse	BioLegend	300512
CD8	RPA-T8	Mouse	eBioscience	47-0088-42
CD11c	3.9	Mouse	BioLegend	301628
SEMA4A	5E3/SEMA4A	Mouse	BioLegend	148404
CD3ε	145-2C11	Armenian Hamster	BioLegend	100328
CD4	GK1.5	Rat	BioLegend	100406
CD8	53-6.7	Rat	BioLegend	100714
CD16/32	93	Rat	BioLegend	101301
CD69	H1.2F3	Armenian Hamster	BioLegend	104514
CD103	2E7	Armenian Hamster	BioLegend	121422
TCR Vγ2	UC3-10A6	Armenian Hamster	BioLegend	137705
TCR Vγ3	536	Syrian Hamster	BD Biosciences	743241
TCRγδ	GL3	Armenian Hamster	BioLegend	118124
IFNγ	XMG1.2	Rat	BioLegend	505813
IL-4	11B11	Rat	BD Biosciences	562915
IL-17A	TC11-18H10.1	Rat	BioLegend	506925

Table S4

Antibodies used for flow cytometry analysis

Cytokines	Supplier	Catalog
IL-1β	BioLegend	575102
IL-6	BioLegend	575702
IL-23	BioLegend	589002
TGFβ1	BioLegend	763102

Table S5

Mouse recombinant cytokines

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Reviewer #1 (Public Review):

Summary:

In this study, Kume et al examined the role of the protein Semaphorin 4a in steady-state skin homeostasis and how this relates to skin changes seen in human psoriasis and imiquimod-induced psoriasis-like disease in mice. The authors found that human psoriatic skin has reduced expression of Sema4a in the epidermis. While Sema4a has been shown to drive inflammatory activation in different immune populations, this finding suggested Sema4a might be important for negatively regulating Th17 inflammation in the skin. The authors go on to show that Sema4a knockout mice have skin changes in key keratinocyte genes, increased gdT cells, and increased IL-17 similar to differences seen in non-lesional psoriatic skin, and that bone marrow chimera mice with WT immune cells and Sema4a KO stromal cells develop worse IMQ-induced psoriasis-like disease, further linking expression of Sema4a in the skin to maintaining skin homeostasis. The authors next studied downstream pathways that might mediate the homeostatic effects of Sema4a, focusing on mTOR given its known role in keratinocyte function. As with the immune phenotypes, Sema4a KO mice had increased mTOR activation in the epidermis in a similar pattern to mTOR activation noted in non-lesional psoriatic skin. The authors next targeted the mTOR pathway and showed rapamycin could reverse some of the psoriasis-like skin changes in Sema4a KO mice, confirming the role of increased mTOR in contributing to the observed skin phenotype.

Strengths:

The most interesting finding is the tissue-specific role for Sema4a, where it has previously been considered to play a mostly pro-inflammatory role in immune cells, this study shows that when expressed by keratinocytes, Sema4a plays a homeostatic role that when missing leads to the development of psoriasis-like skin changes. This has important implications in terms of targeting Sema4a pharmacologically. It also may yield a novel mouse model to study mechanisms of psoriasis development in mice separate from the commonly used IMQ model. The included experiments are well-controlled and executed rigorously.

Weaknesses:

A weakness of the study is the lack of tissue-specific Sema4a knockout mice (e.g. in keratinocytes only). The authors did use bone marrow chimeras, but only in one experiment. This work implies that psoriasis may represent a Sema4a-deficient state in the epidermal cells, while the same might not be true for immune cells. Indeed, in their analysis of non-lesional psoriasis skin, Sema4a was not significantly decreased compared to control skin, possibly due to compensatory increased Sema4a from other cell types. Unbiased RNA-seq of Sema4a KO mouse skin for comparison to non-lesional skin might identify other similarities besides mTOR signaling. Indeed, targeting mTOR with rapamycin reveres some of the skin changes in Sema4a KO mice, but not skin thickness, so other pathways impacted by Sema4a may be better targets if they could be identified. Utilizing WT□KO chimeras in addition to global KO mice in the experiments in Figures 6-8 would more strongly implicate the separate role of Sema4a in skin vs immune cell populations and might more closely mimic non-lesional psoriasis skin.

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Reviewer #2 (Public Review):

Summary:

Kume et al. found for the first time that Semaphorin 4A (Sema4A) was downregulated in both mRNA and protein levels in L and NL keratinocytes of psoriasis patients compared to control keratinocytes. In peripheral blood, they found that Sema4A is not only expressed in keratinocytes but is also upregulated in hematopoietic cells such as lymphocytes and monocytes in the blood of psoriasis patients. They investigated how the down-regulation of Sema4A expression in psoriatic epidermal cells affects the immunological inflammation of psoriasis by using a psoriasis mice model in which Sema4A KO mice were treated with IMQ. Kume et al. hypothesized that down-regulation of Sema4A expression in keratinocytes might be responsible for the augmentation of psoriasis inflammation. Using bone marrow chimeric mice, Kume et al. showed that KO of Sema4A in non-hematopoietic cells was responsible for the enhanced inflammation in psoriasis. The expression of CCL20, TNF, IL-17, and mTOR was upregulated in the Sema4AKO epidermis compared to the WT epidermis, and the infiltration of IL-17-producing T cells was also enhanced.

Strengths:

Decreased Sema4A expression may be involved in psoriasis exacerbation through epidermal proliferation and enhanced infiltration of Th17 cells, which helps understand psoriasis immunopathogenesis.

Weaknesses:

The mechanism by which decreased Sema4A expression may exacerbate psoriasis is unclear as yet.

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